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### HOMWORK 3 (MODERN OPTIMIZATION ALGORITHMS, MATH 485)

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**Due date:** See class webpage on Canvas.

Unless otherwise stated,  $\mathcal{H}$  is a real Hilbert space with inner product  $\langle \cdot | \cdot \rangle$  and induced norm  $\| \cdot \|$ .

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#### QUESTIONS

1. Let

$$x = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 77 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \quad \text{and} \quad k = \begin{pmatrix} \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \end{pmatrix}. \quad (1)$$

Provide the centralmost  $4 \times 4$  “image” resulting from the convolution  $x * k$  with replicated boundary conditions. You may use software for this problem and/or check by hand. If you use software, please print out the code you used for this problem.

2. In class, we learned that if  $C \subset \mathcal{H}$  is nonempty, closed, and convex, then existence and uniqueness of  $\text{Proj}_C(x)$  is guaranteed. This question pertains to what happens if we drop either convexity or closedness of  $C$ .

- (a) Provide an instance of  $\mathcal{H}$ , nonempty  $C \subset \mathcal{H}$ , and  $x \in \mathcal{H}$ , such that  $C$  is convex, and  $\text{Proj}_C(x)$  does not exist. Provide justification of convexity of  $C$  and some viable argument as to why  $\text{Proj}_C(x)$  does not exist.
- (b) Provide an instance of  $\mathcal{H}$ , nonempty  $C \subset \mathcal{H}$ , and  $x \in \mathcal{H}$ , such that  $C$  is closed, and  $\text{Proj}_C(x)$  is not unique. You can provide a justification of non-uniqueness by providing at least two projections. Please explain why the purported projections are projections.

3. Suppose that  $\| \cdot \|_*$  is an arbitrary (potentially non-Hilbertian) norm. Prove  $\| \cdot \|_*$  is convex using the 3 properties of the norm.

*hint:* Using the proposition from class, it would suffice to show

$$(\forall x, y \in \mathcal{H})(\forall \alpha \in ]0, 1[) \quad \|\alpha x + (1 - \alpha)y\|_* \leq \alpha\|x\|_* + (1 - \alpha)\|y\|_*. \quad (2)$$

4. Let  $\alpha, \beta \in \mathbb{R}$  be such that  $\alpha \leq \beta$ , and consider the nonempty set  $C = [\alpha, \beta]$ .

(a) Prove that

$$\text{Proj}_C: \mathbb{R} \rightarrow \mathbb{R}: x \mapsto \begin{cases} \beta & \text{if } x > \beta \\ x & \text{if } \beta \geq x \geq \alpha \\ \alpha & \text{if } \alpha > x \end{cases} \quad (3)$$

(b) Provide a short justification that  $\text{Proj}_C(x) = \underset{c \in C}{\operatorname{argmin}} \|x - c\|^2$ .

*hint: cite a previous homework assignment.*

(c) Consider the hypercube  $H = [\alpha, \beta]^n \subset \mathbb{R}^n$ , i.e.,

$$H = [\alpha, \beta]^n = \{x \in \mathcal{H} \mid (\forall i \in \{1, \dots, n\}) \quad x_i \in [\alpha, \beta]\}. \quad (4)$$

You may assume that  $H$  is nonempty, closed, and convex. Prove that  $(\forall x \in \mathbb{R}^n)$ ,  $\text{Proj}_H(x) = (\text{Proj}_C(x_i))_{i=1}^n$ .